

Robot Central Management System

User Documentation

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Document Preface

This document is intended for system users. It provides a complete introduction to Robot Central all functional modules of the Management System, standard operating procedures, status descriptions, and common troubleshooting solutions. It helps operators quickly get started with the full lifecycle management of autonomous mobile robots and improves the operation, maintenance, and scheduling efficiency of delivery robots.

1. System Overview

It is an integrated comprehensive management and control platform designed for Autonomous Mobile Robots (AMR). It integrates six core capabilities: device ledger, real-time monitoring, remote control, navigation maps, delivery tasks, and call box hardware, supporting full lifecycle visual management of robots.

1.1 Core System Capabilities

- Device Ledger Management: Add, edit, and delete robots; maintain basic device information
- Real-time Status Monitoring: Card-style display of device operating status, battery level, location, and other information
- Remote Control and Scheduling: Issue navigation, charging, pause, jack, and other control commands
- Navigation Map Management: Maintain multi-floor, multi-area navigation map data

- Delivery Task Management: Create, view, and manage the full delivery task workflow
- Call Box Configuration: Manage hardware call boxes and button function binding

1.2 System Menu Structure

The left navigation menu of the system contains the following main functional modules:

Menu	Function Description
Device List	Manage basic information of all AMR devices
Device Status	Monitor device operating status in real-time and issue control commands
Map Management	Manage navigation maps used by devices
Task Management	View and manage delivery tasks
Call Box Configuration	Manage call box devices and their button functions



Figure: System Main Interface and Menu Structure Diagram

2. Device List Management

2.1 View Device List

Entry: Left Menu → Device Management → Device List

After entering the device list page, the system will automatically load all registered device information and display it in a table format.

List Field Description

Field	Note
Device ID	System-generated unique identifier
Device Name	Unique device identifier, format: AMR + number (e.g., AMR001, AMR002)
Device Alias	Custom alias for easy identification (e.g., "Pharmacy Delivery Robot")
Device IP	Network IP address of the device (e.g., 192.168.1.100)
Device Category	Device model classification (e.g., AMR-GT, AMR-BZ)
Department Name	Department the device belongs to
Device Status	"Normal" or "Disabled"
Created Time	Device registration time

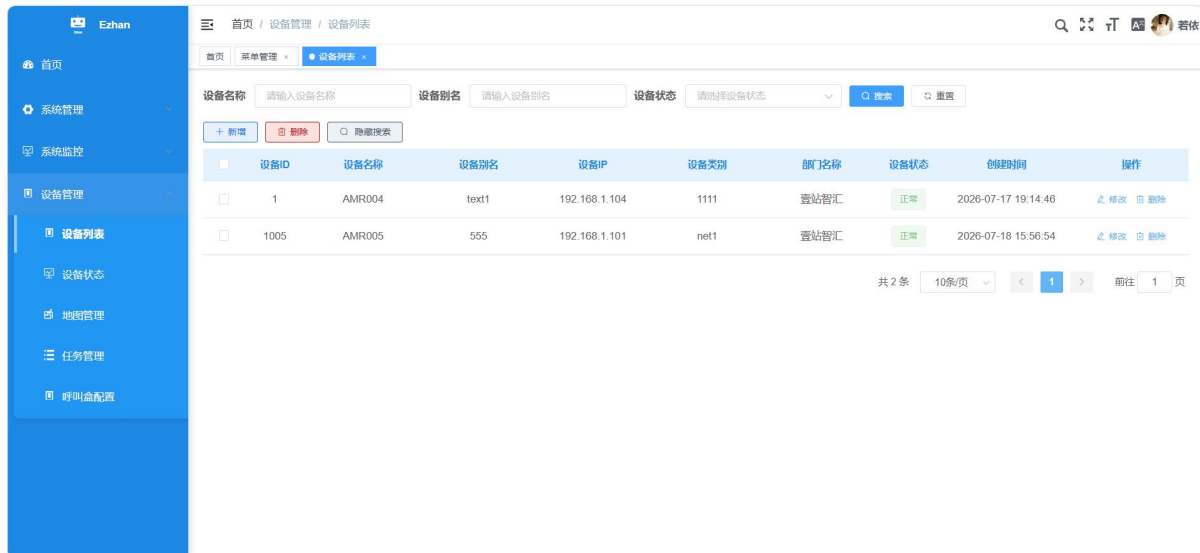


Figure: Device List Page

2.2 Search Devices

In the search area at the top of the device list page, you can filter devices by the following criteria:

- Device Name: Enter device name keywords for fuzzy search
- Device Alias: Enter alias keywords for fuzzy search
- Device Status: Select "Normal" or "Disabled" from the dropdown

Click the "Search" button to execute the filter. Click the "Reset" button to clear search conditions.

2.3 Add Device

Steps:

1. Click the "Add" button above the device list
2. Fill in the device information in the popup form
3. Click the "OK" button to save

Add Device Form Fields

Field	Required	Fill Instructions
Device Name	Required	Format must be AMR + number, e.g., AMR001, AMR010
Device Alias	Optional	Custom device name for easy daily identification
Device IP	Optional	Device IP address, format: 192.168.1.100. The system automatically validates IP format and does not allow duplicates
Device Category	Optional	Device model, e.g., AMR-GT (Cabinet Robot), AMR-BZ (Standard Robot)
Department	Required	Select the department the device belongs to from the dropdown
Device Status	Required	Select "Normal" or "Disabled", default is "Normal"
Remarks	Optional	Additional description information for the device

Tip

Device IP addresses cannot be duplicated. If the entered IP is already used by another device, the system will show an error.



Figure: Add Device Dialog

2.4 Edit Device

1. In the device list, click the "Edit" button in the target device row
2. Modify the information that needs to be changed in the popup form
3. Click the "OK" button to save changes

Note

The device ID in the device name cannot be modified.

2.5 Delete Device

- Single Delete: Click the "Delete" button in the target device row
- Batch Delete: Check the checkboxes before multiple devices, then click the "Delete" button above the list

A confirmation dialog will pop up before deletion. Please confirm before executing the delete operation.

Warning

Deleted devices cannot be recovered. Please proceed with caution.

3. Device Status Monitoring

3.1 Device Status Overview

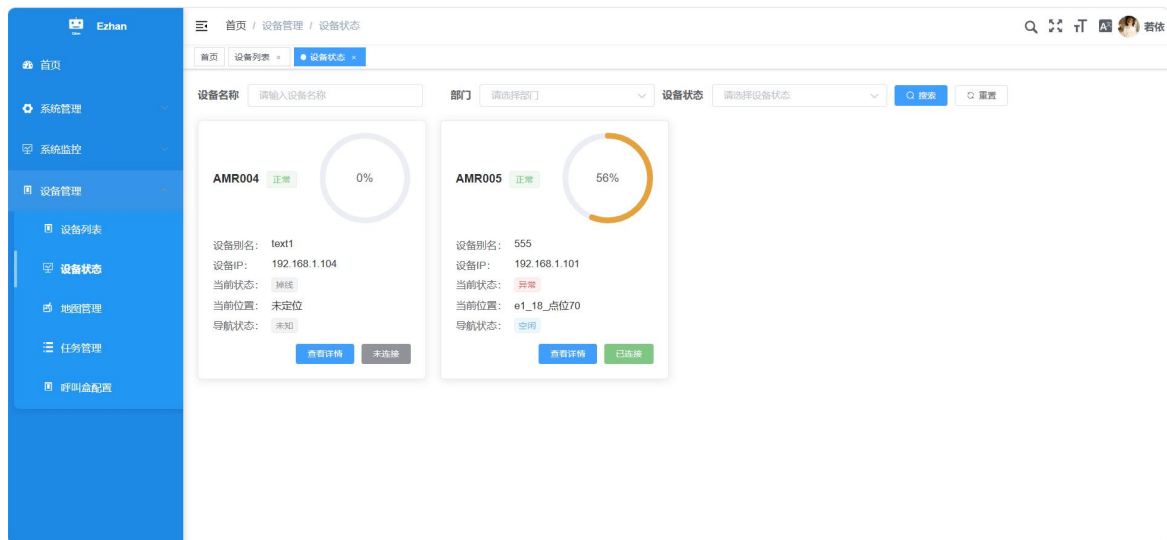
Entry: Left Menu → Device Management → Device Status

The device status page displays the real-time operating status of all devices in a card view. It is the core page for daily monitoring. Each device is displayed as a card.

Device Card Information

Information Item	Note
Device Name	Unique device identifier (e.g., AMR001)
Device Status	Green "Normal" or Red "Disabled"
Battery Ring	Display current battery level percentage with a ring progress bar
Device Alias	Custom device name
Device IP	Device network address
Current Status	Device operating status
Current Location	Station name or coordinates where the device is currently located
Navigation Status	Device navigation execution status
Connection Status	Bottom button displays connection status

Figure: Device Status Card Views follows



3.2 Device Operating Status Description

Status	Color	Note
Idle	Blue	Device is online, not currently executing any task
Running	Green	Device is executing a delivery task
Abnormal	Red	Device has an anomaly (emergency stop, collision bar triggered, etc.)
Offline	Gray	Device cannot be connected, may be powered off or experiencing network interruption

3.3 Navigation Status Description

Status	Note
Idle	Device is not currently navigating
Running	Device is navigating to the target position
Completed	Device has reached the target position
Navigation Failed	An issue occurred during navigation, device failed to reach target

3.4 Battery Level Display Rules

Battery Range	Color	Suggested Action
70% ~ 100%	Green	Sufficient battery, normal use
30% ~ 70%	Yellow	Medium battery, monitor as needed
0% ~ 30%	Red	Low battery, schedule charging as soon as possible

3.5 Connect Device

Before using the remote control function, you need to connect the device first:

1. Click the "Disconnected" button at the bottom of the device card
2. The system starts connecting the device, the button displays "Connecting"
3. After successful connection, the button turns green "Connected", and the card will display real-time status information of the device

Note

The system will automatically refresh the status information of all connected devices every 2 seconds.

3.6 View Device Details

You can view device details in the following two ways:

- Method 1: Double-click the device card
- Method 2: Click the "View Details" button at the bottom of the card

The details page is divided into four information areas:

Basic Information

Field	Note
Device Name	Unique device identifier
Device Alias	Custom Alias
Device IP	Network Address
Device Category	Device Model
Department	Department
Device Status	Normal/Disabled
Created Time	Registration Time

Operating Status

Field	Note
Operating Status	Idle/Running/Abnormal/Offline
Emergency Stop Status	Whether the emergency stop button or soft stop has been triggered
Wheel Status	Whether the drive wheels are locked
Collision Bar	Whether the collision protection bar has been triggered

Runtime Information

Field	Note
Battery Level	Current battery level percentage (with progress bar)
Mileage	Device cumulative mileage (unit: km)
Running Time	Device cumulative running time (unit: hours)
Current Location	Station name or coordinates where the device is located
Current Map	Name of the navigation map currently used by the device
Navigation Status	Current navigation execution status of the device
Positioning Confidence	Accuracy percentage of device autonomous positioning

System Information

Field	Note
System Version	Version number of the device control system
Last Maintenance	Time of last maintenance
MAC Address	Network hardware address of the device

Note

After opening the details page, the system will automatically refresh the device status information every 1 second to ensure you see the latest data.



Figure: Device Details Page (Note: Go to Pharmacy/IV Compounding Room/Emergency Pharmacy are medical version robot functions)

4. Remote Device Control

At the bottom of the device details page, you can issue various control commands to connected devices. All commands require confirmation before execution.

4.1 Go to Specified Location

Steps:

1. Click the corresponding button (e.g., "Go to Pharmacy")
2. Click "OK" in the confirmation dialog
3. The system sends a navigation task to the device via WebSocket
4. Display a prompt message after success

Common Navigation Buttons

Button	Function Description
Go to Standby Point	Move the device to the preset standby/rest position
Go to Pharmacy	Move the device to the pharmacy position
Go to IV Compounding Room	Move the device to the IV compounding room position
Go to Emergency Pharmacy	Move the device to the emergency pharmacy position

4.2 Charging Task

Click the "Charging Task" button, and the device will automatically go to the charging area to charge.

4.3 Pause and Resume Task

Button	Function Description
Pause Task	Pause the task currently being executed by the device. The device will stop in place
Resume Task	Resume the previously paused task. The device continues execution

Typical Usage Scenario

When there are temporary obstacles on the device path, you can pause the task first and resume it after the obstacles are cleared.

4.4 Cancel Task

Click the "Cancel Task" button, and the device will stop all currently executing tasks.

Note

After canceling the task, the device will stop at the current position and will not automatically return to the standby point. If you need to return, please manually issue the "Go to Standby Point" command.

4.5 Jack Task

Click the "Jack Task" button to issue a jack action command to the device (applicable to devices with jack mechanisms).

4.6 Remote Control (Requires 4G)

Click the "Remote Control" button, and the system will automatically open a new browser tab to enter the device remote control interface.

In this interface, you can manually control the movement direction and speed of the device.

4.7 Search and Filter

The top of the page provides a search area where you can filter devices by the following criteria:

- Device Name: Enter name keywords
- Department: Select department from the dropdown
- Device Status: Select "Normal" or "Disabled"

5. Map Management

5.1 View Map List

Entry: Left Menu → Device Management → Map Management

Map Management is used to maintain navigation map data for AMR devices. Devices need to load the correct map to perform autonomous navigation.

List Field Description

Field	Note
Map ID	System-generated map number
Map Name	Map name identifier
Department	Department the map belongs to
Status Code	Map enable status
Created Time	Map upload time

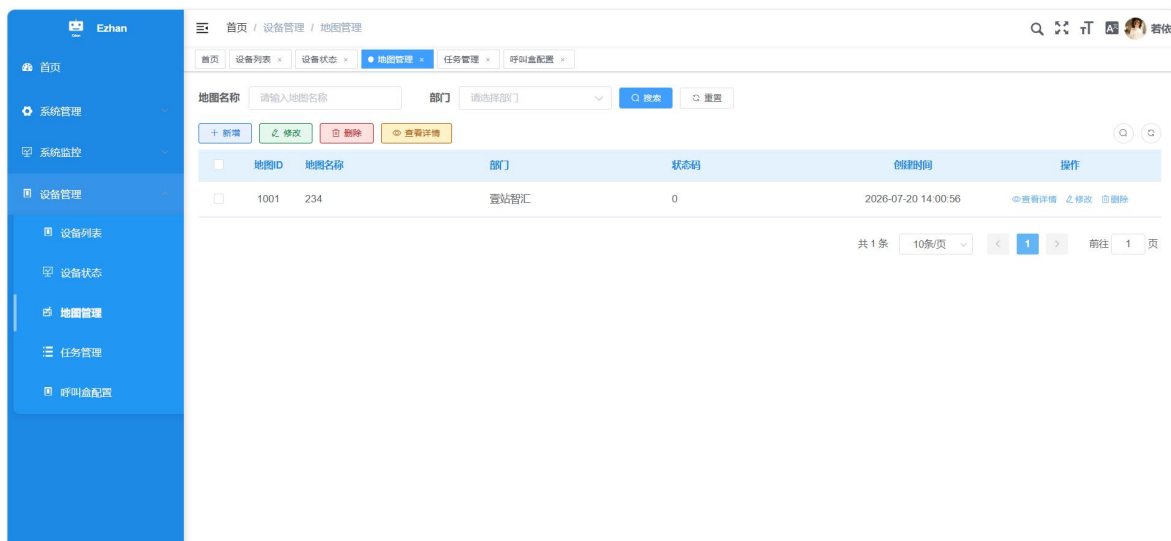


Figure: Map Management List Page

5.2 Add Map

1. Click the "Add" button

2. Fill in the map information
3. Click "OK" to save

Add Map Form

Field	Required	Note
Map Name	Required	Map name, use meaningful names (e.g., "First Floor Pharmacy Area")
Department	Required	Department the map belongs to
Map Data	Optional	Map JSON data, typically imported from device-generated map data
Status Code	Optional	Select "Enabled" or "Disabled"
Remarks	Optional	Additional description for the map

5.3 View Map Details

Click the "View" button in the map row to view detailed map information, including map image preview.

5.4 Edit and Delete

- Edit: Click the "Edit" button to edit the map name, department, data, and other information
- Delete: Click the "Delete" button to delete the selected map (supports batch deletion)

Note

Deleting a map in use may affect the normal navigation of devices. Please confirm no devices are using it before deleting.

6. Task Management

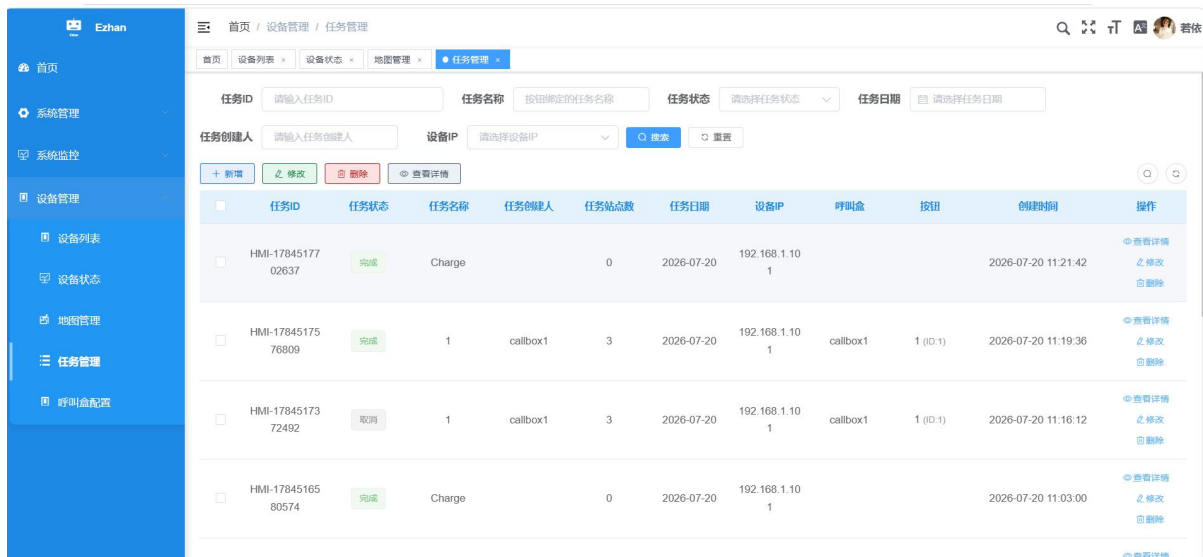
6.1 View Task List

Entry: Left Menu → Device Management → Task Management

The task management page is used to view and manage delivery task records of all AMR devices.

List Field Description

Field	Note
Task ID	Unique task identifier
Task Status	Current task status
Task Name	Task name description
Task Creator	Operator who created the task
Task Station Count	Number of stations the task needs to pass through
Task Date	Task execution date
Device IP	IP address of the device executing the task
Call Box	Name of the call box that initiated the task
Button	Name of the button that triggered the task
Created Time	Task creation time



任务ID	任务状态	任务名称	任务创建人	任务站点数	任务日期	设备IP	呼叫盒	按钮	创建时间	操作
HMI-1784517702637	完成	Charge		0	2026-07-20	192.168.1.101			2026-07-20 11:21:42	查看详情 修改 删除
HMI-1784517576809	完成	1	callbox1	3	2026-07-20	192.168.1.101	callbox1	1 (ID:1)	2026-07-20 11:19:36	查看详情 修改 删除
HMI-1784517372492	取消	1	callbox1	3	2026-07-20	192.168.1.101	callbox1	1 (ID:1)	2026-07-20 11:16:12	查看详情 修改 删除
HMI-1784516580574	完成	Charge		0	2026-07-20	192.168.1.101			2026-07-20 11:03:00	查看详情 修改 删除

Figure: Task Management List Page

6.2 Task Status Description

Status	Color Label	Note
Pending	Gray	Task has been created but not yet dispatched to device
Running	Orange	Task has been dispatched, device is executing
In Progress	Orange	Task is currently being executed
Completed	Green	Task has been successfully completed
Cancelled	Red	Task has been cancelled
Not Started	Gray	Task has not started execution

Note

The task list automatically refreshes every 3 seconds to ensure timely status updates.

6.3 Search Tasks

Supports the following search criteria:

Condition	Note
Task ID	Exact search by task number
Task Name	Fuzzy search by name keywords
Task Status	Select specific status from dropdown
Task Date	Select specific date
Task Creator	Search by creator name
Device IP	Select device executing the task from dropdown

6.4 Add Task

1. Click the "Add" button
2. Fill in the task information
3. Click "OK" to save

Add Task Form

Field	Required	Note
Task ID	Required	Unique task identifier, cannot contain spaces, no more than 50 characters
Task Status	Required	Select the initial task status
Task Name	Optional	Task description name
Task Creator	Optional	Person who created the task
Task Station Count	Optional	Number of stations for delivery (positive integer)
Task Date	Optional	Task execution date (cannot be earlier than today)
Device IP	Optional	Device to execute the task

6.5 View Task Details

Click the "View" button in the task row to view the station details of the task.

Cabinet Robot (AMR-GT) Station Details

Field	Note
Station Name	Delivery station name
Arrival Time	Time when the device arrived at the station
Door Open Time	Time when the cabinet door opened
Door Opener	Person who opened the cabinet door
Door Close Time	Time when the cabinet door closed
Departure Time	Time when the device left the station
Door Number	Number of the opened cabinet door

Standard Robot (AMR-BZ) Station Details

Field	Note
Station Name	Station name
Stay Duration	Time the device stayed at the station
Action Type	Action executed by the device (move, jack, place, etc.)
Current Loop	Number of loops currently executed
Total Loop Count	Total number of loops to execute
Target Station	Current navigation target station

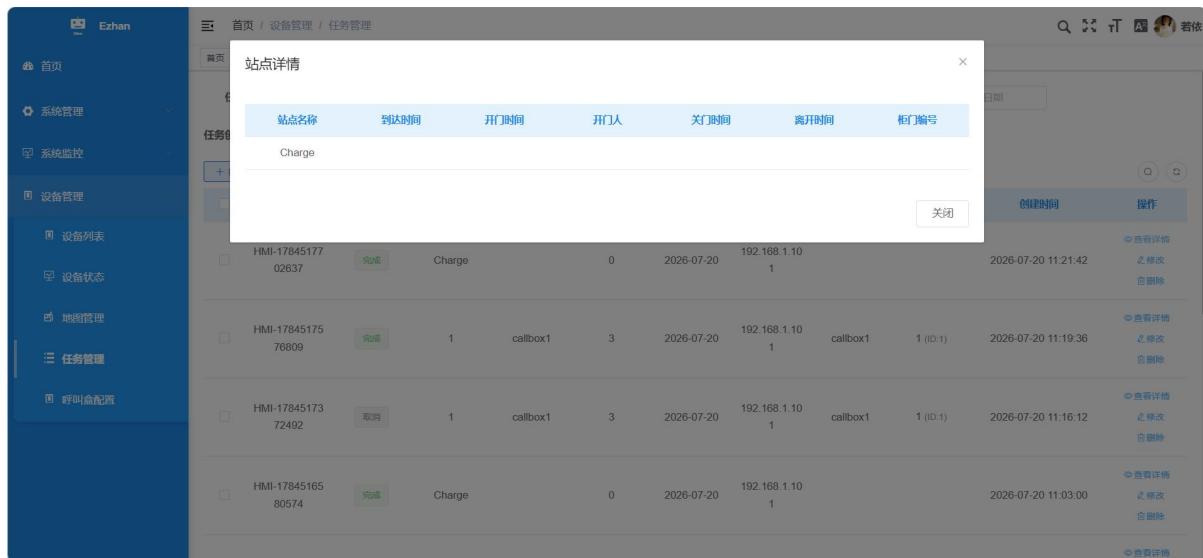


Figure: Task Details Page

6.6 Edit and Delete

- Edit: Click the "Edit" button to edit the task status, name, and other information
- Delete: Select the task and click the "Delete" button (supports batch deletion)

Tip

Deleting a task will not cancel the command being executed by the device. If you need to cancel the device action, please go to the "Device Status" page to issue a cancel task command.

7. Call Box Configuration

7.1 Call Box List

Entry: Left Menu → Device Management → Call Box Configuration

List Field Description

Field	Note
ID	System-generated number
Call Box Name	Call box name (e.g., "First Floor Nurse Station Call Box")
Device IP	Call box network address
ICCID/IP	SIM card ICCID number or IP address of the call box
Department Name	Department the call box belongs to
Device Status	"Online" or "Offline"
Bound Task Name	Default task name executed when button is pressed
Auto Assign	Whether to enable auto-assign idle robot function
Button Count	Number of buttons configured for the call box
Created Time	Call box registration time

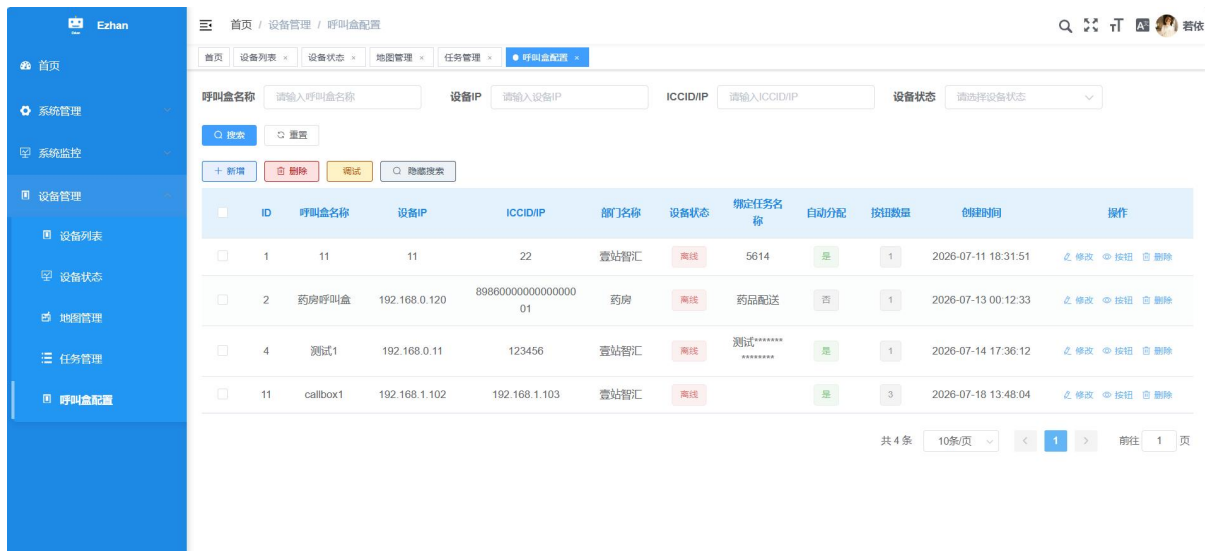


Figure: Call Box Configuration List Page

7.2 Add Call Box

1. Click the "Add" button
2. Fill in basic information
3. Configure button functions (see below)
4. Click "OK" to save

Add Call Box Form

Field	Required	Note
Call Box Name	Required	Call box name, include location information
Device IP	Required	Network IP address of the call box
ICCID/IP	Required	SIM card ICCID number or IP address of the call box
Department	Required	Department the call box belongs to
Bound Task Name	Optional	Default task name executed when button is pressed
Auto Assign	Optional	When enabled, the system will automatically assign the task to an idle robot in the same department when the

Field	Required	Note
		button is pressed
Device Status	Optional	Select "Online" or "Offline", default is offline
Remarks	Optional	Additional description for the call box

7.3 Button Configuration

Each call box can configure multiple physical buttons, and each button can bind different tasks and stations. Find the "Button Configuration" area in the add/edit call box form.

Button Configuration Fields

Field	Note
Button ID	Physical button number (numeric)
Button Name	Button name identifier (e.g., "Call Delivery", "Emergency Call")
Bound Task	Task name executed when the button is pressed
Bound Station	Target station associated with the button, multiple stations separated by commas

Operation Instructions:

- Click "Add Button" to add a new button configuration row
- Click the delete icon on the right side of the row to remove the button configuration



Figure: Call Box Button Configuration Interface

7.4 View Button Configuration

Click the "Button" button in the call box list to view all buttons currently configured for the call box and their binding information.

7.5 Edit and Delete

- Edit: Click the "Edit" button to edit all information and button configuration of the call box
- Delete: Click the "Delete" button to delete the selected call box (supports batch deletion)

7.6 Debug Function

The debug function is used to view communication logs between the call box and the system, helping to troubleshoot call box issues.

Steps:

1. Click the "Debug" button above the list

2. Enter the debug password for authentication
3. Select the call box to debug:
 - Click "Debug All Call Boxes" to view logs of all call boxes
 - Click a row in the table to view logs of a single call box

Debug Page Functions

Function	Note
Auto Refresh	Automatically load latest logs every 2 seconds when enabled
Pause Refresh	Stop auto refresh
Clear Log	Clear all currently displayed log records
Export Log	Export logs as JSON file to local storage
Filter Type	Filter by log type: All/Heartbeat/Button Call/Response/Process

Log Type Description

Type	Note
Heartbeat	Heartbeat signal periodically sent by the call box, indicating the device is online
Button Call	Call request sent when the call box button is pressed
Response	System reply to the call request
Process	Internal process record of the system processing the call request

Log Information Description

Each log entry contains the following information:

- Time: Log recording time
- Type: Log type (Heartbeat/Call/Response/Process)

- Direction: "Received" indicates message from call box, "Sent" indicates message from system to call box
- Parsed Information: System interpretation of the original message
- Original Message: Original data transmitted between the call box and the system
- Error Information: If an error occurs during processing, error details will be displayed here

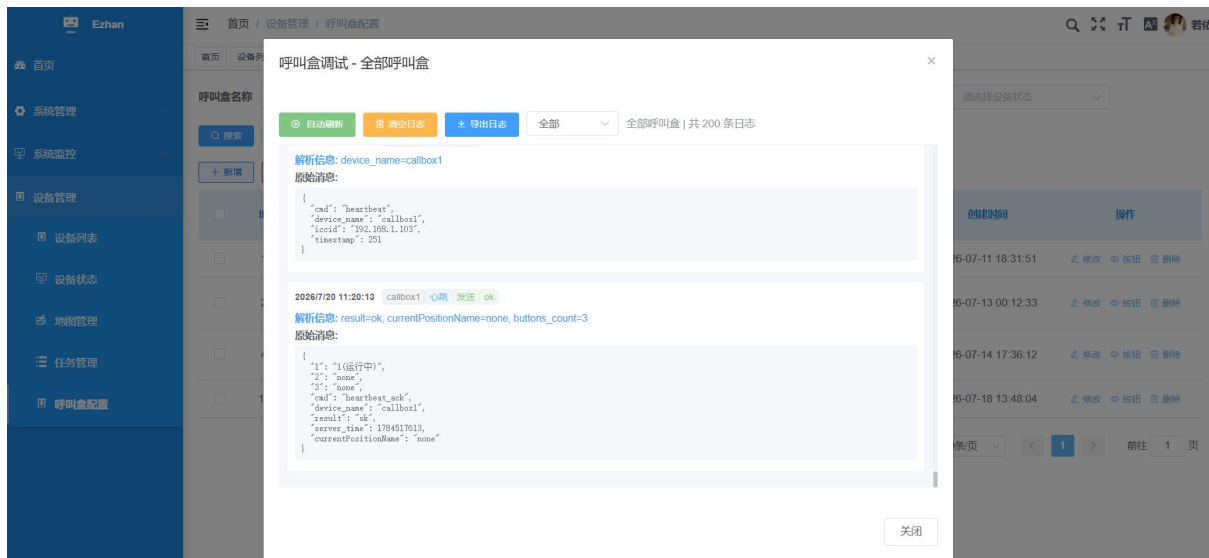


Figure: Call Box Debug Log Page

8. FAQ and Notes

8.1 Device Connection

Q: What should I do if the device shows "Offline" status?

A: Please check the following in order:

- Whether the device is powered on
- Whether the device network connection is normal
- Whether the device IP address is configured correctly
- Click the "Connect" button to try reconnecting

Q: What does "Connection Failed" mean when connecting a device?

A: Possible reasons:

- Incorrect device IP address entered
- Device is not on the same network
- Device is being operated by someone else
- Network delay is too high, please try again later

8.2 Task Control

Q: What should I do if the device doesn't respond after issuing a task?

A: Please check:

- Whether the device is connected (status should show "Connected")
- Whether the device is in an abnormal state (emergency stop, collision bar triggered, etc.)
- Whether the device already has a task being executed

- Whether the network connection is normal

Q: How to make the device return to the pharmacy from the current position?

A: On the device details page, click the "Go to Pharmacy" button.

Q: What should I do if the device encounters an obstacle during task execution?

A: You can click "Pause Task" to stop the device first, and click "Resume Task" to continue execution after the obstacle is cleared.

8.3 Call Box

Q: What should I do if no robot responds after pressing the call box button?

A: Please check:

- Whether the call box device status is "Online"
- Whether the call box button configuration is correct (bound task, bound station)
- Whether there are idle robots available
- Whether the call box "Auto Assign" function is enabled
- You can use the debug function to view the call box communication logs for troubleshooting

Q: How to check if the call box is working properly?

A: Use the debug function, enable auto refresh, and observe whether heartbeat logs appear periodically. If there are no heartbeat logs for a long time, the call box may be offline or experiencing network issues.

8.4 General Notes

- Operation Confirmation: All control commands (go to location, charging, pause, cancel, etc.) will pop up a confirmation dialog before execution. Please confirm carefully before operating
- Device Name Convention: Device names must start with AMR followed by a number (e.g., AMR001). This is the system naming convention
- IP Address Uniqueness: Each device IP address must be unique and cannot be duplicated
- Permission Management: Different functions require different operation permissions. If you don't see certain buttons or menus, please contact the administrator to assign the corresponding permissions
- Data Security: Deleting devices, maps, tasks, etc. cannot be recovered. Please confirm before deletion
- Real-time Refresh: Device status page auto-refreshes every 2 seconds, task list auto-refreshes every 3 seconds. No need to manually refresh the page
- Browser Recommendation: Chrome, Edge, and other modern browsers are recommended for the best experience

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